

# Sarveswaran MG

## Software Engineer

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Software Engineer with experience designing, developing, and debugging scalable web and AI-driven applications. Strong foundation in software architecture, Data Structures and Algorithms, and backend development. Skilled in building high-performance systems using modern frameworks, cloud-native platforms, and LLM integrations, with a focus on reliability, scalability, and user experience.

## EDUCATION

**M.Tech (Integrated) | CSE with Specialization in Business Analytics** 2020 - 2025  
Vellore Institute of Technology, Chennai Campus CGPA: 8.77

**Higher Secondary** 2019 - 2020  
C.E.O.A Matriculation Higher Secondary School, Madurai 85.2 %

## SKILLS

- Programming Languages: **Python, JavaScript, Java, SQL, HTML, CSS.**
- Frontend: **React, Redux, Tailwind CSS, GSAP, Vite**
- Backend & APIs: **FastAPI, Node.js, REST APIs, Webhooks, Celery**
- Cloud & DevOps: **AWS (S3, Lambda), Firebase, Vercel, CI/CD Pipelines, Git**
- AI & Data: **LLMs, LangChain, Prompt Engineering, Pandas, NumPy, XGBoost**
- Testing: **Jest, Unit Testing, Integration Testing**
- Databases: **MongoDB, Elasticsearch, DynamoDB, Firebase Firestore**

## EXPERIENCE

**SOFTWARE INTERN – Big Think Code** March 2026 - Present

- Architecting Shelf Analytics, a cloud-native AI retail analytics platform integrating LLM workflows via **LangChain** for product detection and shelf intelligence, deployed on **AWS S3 and Lambda**.
- Built scalable **FastAPI microservices** and asynchronous image-processing pipelines; troubleshoot and debugged pipeline failures, improving system reliability in production; deployed via CI/CD pipelines on **AWS, leveraging S3 for storage and Lambda for serverless compute**.
- Researched and evaluated **Elasticsearch** for semantic search and product indexing capabilities as part of the CulinaryBot project, assessing its fit for **scalable recipe and ingredient search workflows**.

**FRONTEND DEVELOPER (Freelance) – NOCTURNAL COMPANY** September 2025 – March 2026

- **ANFED** 📁: Built a geospatial agriculture analytics platform using React, achieving **60–80% rendering performance** improvement via **lazy loading, code splitting, and optimized data visualizations**.
- **POROUS BEING** 📁: Delivered a responsive, accessible (WCAG-aligned), **SEO-optimized portfolio** with **GSAP** animations and modular component architecture, optimized for Core Web Vitals.
- **FENIVI** 📁: Developed a performant **SPA** with real-time Firestore subscriptions, animated **UI using Tailwind CSS**, client-side routing, and scalable component-based design patterns.

**FULL STACK DEVELOPER INTERN - ADRIG AI** July 2025 – December 2025

- Developed responsive, accessible UI components from **Figma designs using React**, maintaining **cross-browser compatibility** and design fidelity.
- Supported **RESTful API development** and **webhook** integrations for system-to-system communication within an Agile sprint environment.

**FULL-STACK DEVELOPER (Freelance) – RENTO** March 2025 – August 2025

- Built a scalable **multi-tenant cab-booking** platform with RBAC, live map integration, payment processing, and ~35% performance improvement via **lazy loading** and **code splitting**.
- Developed **REST APIs** and integrated third-party services supporting real-time operations across multiple user roles.
- Implemented **Jest-based unit** and integration tests for critical booking and payment workflows.

## CERTIFICATIONS

[Google Cloud Computing Foundation Program](#) 📁, [Foundations of User Experience \(UX\) Design](#) 📁, [Entrepreneurship](#) 📁, [Intellectual Property](#) 📁, [React Fundamentals](#) 📁, [React Hooks](#) 📁